

This project is to direct you to the basic working concept of radio interferometry using the Submillimeter Array (SMA) atop Mauna Kea in Hawaii. The array consists of 8 antennas each with a diameter $d = 6$ meter, operating at the wavelengths $\lambda = 1.3 - 0.43$ mm. It has 5 different configurations: subcompact, compact, compact-north, extended, and very extended. In this project, only the compact, extended and very extended configurations are used, with their coordinates attached in the right-handed equatorial system.

1st Part (70%)

Calculate the values of u , v , and w in $k\lambda$ at $\lambda = 0.85$ mm in the 3 configurations for sources at 4 different declinations $\delta = -30^\circ, 0^\circ, 30^\circ$ and 60° with hour angles $H = -4.5$ to 4.5 hr. Plot the uv coverages and explain the different uv structures in terms of declinations and configurations. What are the maximum values of uv distance and w in each case, where the uv distance is given by $\sqrt{u^2 + v^2}$?

Assume that the reflector of each antenna has a Gaussian illumination with a power taper of 20 dB at the edge, calculate the primary beam pattern A and the half power beam width (HPBW). Calculate the fringe visibilities

$$\mathcal{V}(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{A(\xi, \eta) B(\xi, \eta)}{\sqrt{1 - \xi^2 - \eta^2}} e^{-i2\pi(u\xi + v\eta)} d\xi d\eta$$

with Fourier Transform at $\lambda = 0.85$ mm in the 3 configurations for the following examples assuming a phase center at $\delta = 30^\circ$ with hour angles $H = -4.5$ to 4.5 hr:

- a. A Gaussian source at the phase center with the following brightness distribution:

$$B(\xi, \eta) = \frac{4 \ln 2}{\pi a^2} \exp[-(4 \ln 2)(\frac{\xi^2}{a^2} + \frac{\eta^2}{a^2})]$$

where $a = 1''$.

- b. Two Gaussian sources with the following brightness distribution:

$$B(\xi, \eta) = \frac{4 \ln 2}{\pi a^2} \exp[-(4 \ln 2)(\frac{(\xi - \xi_1)^2}{a^2} + \frac{\eta^2}{a^2})] + \frac{4 \ln 2}{\pi a^2} \exp[-(4 \ln 2)(\frac{(\xi - \xi_2)^2}{a^2} + \frac{\eta^2}{a^2})]$$

where $a = 1''$, $\xi_1 = 1''$, $\xi_2 = -1''$.

- c. A ring structure with the following brightness distribution:

$$B(\xi, \eta) = 4(\frac{4 \ln 2}{\pi a^2}) \exp[-(4 \ln 2)(\frac{\xi^2}{a^2} + \frac{\eta^2}{a^2})] - \frac{4 \ln 2}{\pi b^2} \exp[-(4 \ln 2)(\frac{\xi^2}{b^2} + \frac{\eta^2}{b^2})]$$

where $a = 2''$ and $b = 1''$.

In each case, compare the visibility amplitude $|\mathcal{V}|$ along the u -axis with the analytical solution and discuss what configuration is needed to spatially resolve the brightness distribution.

2nd Part (50%)

Now, check your results with “MIRIAD”, one of the standard packages used in Radio Interferometry. This package is already set up on a Linux machine (see below) and please login with **ssh -X**. The following programs or routines will be used: **uvgen**, **uvamp**, **uvplt**, **invert**, **clean**, **restor**, and **implot**. For help on e.g., uvgen, type: mirhelp uvgen.

- a. **uvgen**: generate the visibility data
 e.g., `uvgen source=gaussian1 ant=SMA_Compact_nanosec baseunit=+1 telescop=altaz,0 corr=0 time=08APR30:02:16:41.6 freq=345 radec=3:43:56.60,30:00:00.00 harange=-4.5,4.5,0.1 lat=40 pbfwhm=36.0 jyperk=150 out=test.vis systemp=75,290,0.15.`

Here “gaussian1” is a text file for the gaussian source in the example ”a”:

1.0,0.,0.0,1.0,1.0,0,0,0,0

SMA_Compact_nanosec: the text file containing the X Y Z positions of the compact configuration in nanosecond (1 meter $\Rightarrow$ 3.335646 nanosec):

```
-59.7590  -198.6648  100.2996
 17.4407   -66.9682  -49.5127
 -1.6909   -83.9030   4.2720
-21.3492  -226.8298  12.1341
-19.0214   -63.3245  52.0737
  0.0000    0.0000   0.0000
 39.1589  -179.0064 -135.8393
 14.8204  -213.0599 -72.8423
```

- b. **uvamp**: plot the visibility amplitude versus *uv* distance.
 e.g., `uvamp vis=test.vis device=/xw`
- c. **uvplt**: plot the visibility data
 e.g., `uvplt vis=test.vis device=/xw axis=uc,vc options=nobase`
 e.g., `uvplt vis=test.vis device=/xw axis=uvdistance,amp options=nobase`
- d. **invert**: generate dirty beam and map with inverse Fourier Transform
 e.g. `invert vis=test.vis map=test.map beam=test.bm imsize=400 cell=0.06 robust=0 line=wide,1,1,1,1 slope=0.1,interpolate options=systemp sup=0`
- e. **clean**: generate clean model by deconvolving the dirty map with the dirty beam
 e.g., `clean map=test.map beam=test.bm out=test.cln gain=0.05 cutoff=0.01 niters=4000 phat=0.1 minpatch=101 speed=-1. region=quarter`
- f. **restor**: generate the final map by convolving the clean model with the main lobe of the dirty beam.
 e.g., `restor map=test.map beam=test.bm model=test.cln out=test.res`
- g. **implot**: plot the maps and beams
 e.g., `implot in=test.map device=/xw beamquad=3 units=s`
 e.g., `implot in=test.bm device=/xw beamquad=3 units=s`
 e.g., `implot in=test.cln device=/xw beamquad=3 units=s`
 e.g., `implot in=test.res device=/xw region=q beamquad=3 units=s`

Compare the angular size of the main (synthesize) beam in each case with the maximum *uv* distance and explain the comparison. Show the dirty map and dirty beam for each case and discuss them. Show the final maps and discuss them.

Computer: hilo.asiaa.sinica.edu.tw (140.109.177.179). Use **ssh -X** to login

Accounts: st01, st02 ..., st12

Password: To be announced!

Use **passwd** to change your password!

Compact			Extended		
X (m)	Y(m)	Z(m)	X (m)	Y(m)	Z(m)
-17.91528	-59.55813	+30.06901	+26.51830	-133.59591	-104.17710
+5.22859	-20.07652	-14.84352	-51.43460	-136.75912	+107.91910
-0.50691	-25.15346	+1.28072	-42.59819	-71.55192	+86.19645
-6.40032	-68.00175	+3.63770	+11.73992	-53.66413	-40.72546
-5.70245	-18.98418	+15.61129	-11.94082	-178.81342	-0.72140
+0.00000	+0.00000	+0.00000	+0.00000	+0.00000	+0.00000
+11.73953	-53.66469	-40.72353	-6.40008	-68.00231	+3.63888
+4.44305	-63.87365	-21.83755	-16.56271	-27.02429	+30.76939

Very Extended		
X (m)	Y(m)	Z(m)
-131.33806	-251.58774	+309.40974
-37.63612	-349.68630	+31.68535
-42.59809	-71.55184	+86.19660
-91.02546	-479.04143	+145.72300
-128.05029	-73.50657	+295.30118
+0.00000	+0.00000	+0.00000
+26.51720	-133.59640	-104.17513
+11.74086	-53.66402	-40.72575